TOPIC 3: FINANCIAL MANAGEMENT • HIGH-PRIORITY EXAM PRACTICE

Extra Exam Problems: Circonsa, Infopymes, Wifinet, and Cemed

Official Source Document: `T3_Additional_EXERCISES.pdf` (UC3M Aerospace Engineering) • Step-by-step exam strategy, analytical deductions, and balanced financial statements.

Exam Problem 1 • High Difficulty

CIRCONSA — Semester Financial Statements & Cash Reconciliation

Original statement · T3_Additional_EXERCISES.pdf

The financial director of **CIRCONSA** is working on its financial statements (Balance Sheet and Income Statement) as of **30 June 2015 (30/06/2015)**. Starting from the Balance Sheet at 31/12/2014 (Current Assets: 5,400 [Cash 1,200, Receivables 1,800, Stock 2,400]; Non-Current Assets: Machinery 500 less Accum Dep 50 = 450; Current Liabilities: 800 [Taxes 350, Suppliers 450]; Owners' Equity: 5,050 [Retained Earnings 850, Capital 4,200]; Total = 5,850 m.u.), evaluate the semester operations from 01/01/2015 to 30/06/2015:

- **i. Sales:** 1,200 units at 20 mu/u (200 u/month). 30-day payment term. VIP client (30% of sales) receives a 10% commercial discount.
- **ii. Purchases:** 1,000 units (40% in Q1, 60% in Q2) at 12 mu/u. 90-day supplier payment term.
- **iii. Initial Stock:** 200 units at 12 mu/u (2,400 mu).
- **iv. Prior Year Liabilities:** Q1 2015 paid 2014 unpaid taxes (350 mu) and supplier balance (450 mu).
- **v. Prior Year Receivables:** Q1 2015 collected 2014 customer debt except 300 mu deferred to July 2016.
- **vi. Operating Expenses:** Personnel 350 mu/mo + Social Security 150 mu/mo (paid in month produced); Overheads 300 mu/mo (paid 2 months after).
- **vii. Renting:** 1 April 2015 computer renting for 400 mu/month, payment term 120 days.
- **viii. Patent:** 1 April 2015 purchased patent for 2,400 mu, payable 1 July 2015. 4-year linear amortization.
- **ix. Bank Credit:** 1 Feb 2015 borrowed 2,000 mu at 12% per annum, principal + interest due 1 Feb 2016.
- **x. Machine 1:** Bought 1 July 2014, amortized linearly with 20% residual value.
- **xi. Capital & Machine 2:** 1 Feb 2015 cash capital increase 10,000 mu. Same date bought Machine 2 for 12,000 mu (paid 8,000 mu cash, 4,000 mu payable 1 July 2015; 4-year life, 0

residual value).

- **xii. Corporate Tax:** 20% tax, recorded as "taxes to pay" on 30 June 2015.

EXAM DAY STRATEGY: TIME HORIZONS & DEFERRED PAYABLES

- **Semester Duration:** $t = 6$ months $= 0.5$ year. Multiply monthly figures by 6 (or active months).
- **Payable on 1 July 2015:** The patent debt ($2,400$ mu) and Machine 2 debt ($4,000$ mu) are payable *the day after* the balance date. Therefore, they are **Current Liabilities on 30/06/2015**, not cash outflows!
- **Customer Deferral:** The 300 mu deferred to July 2016 is > 1 year away $\implies$ **Non-Current Account Receivable**.
- **Payment Terms:** 30-day customer delay means June sales are unpaid on 30 June; 90-day supplier delay means all Q2 purchases are unpaid on 30 June.

Analytical Deductions Step-by-Step

1. **Net Sales:** Total gross sales $= 1,200 \times 20 = 24,000$ mu. VIP client ($30\% = 7,200$ mu) discount of $10\% = 720$ mu.

$$\text{Net Sales} = 24,000 - 720 = \mathbf{23,280} \text{ mu}$$

   June sales ($1/6$ of total) $= 3,880$ mu are uncollected on 30/06 $\implies$ **Current Accounts Receivable** $= \mathbf{3,880}$ mu. Collections $= 23,280 - 3,880 = \mathbf{19,400}$ mu.

2. **COGS & Ending Inventory:** Available units $= 200$ init $+ 1,000$ purch $= 1,200$ units. Sold $= 1,200$ units $\implies$ **Ending Inventory** $= \mathbf{0}$ mu.

$$\text{COGS} = 1,200 \times 12 = \mathbf{14,400} \text{ mu}$$

   Q2 purchases (600 units $\times 12 = 7,200$ mu) payable at 90 days are unpaid on 30 June $\implies$ **Accounts Payable (Suppliers)** $= \mathbf{7,200}$ mu.

3. **Personnel, Overheads & Renting:**
   - Personnel: 6 months $\times (350 + 150) = \mathbf{3,000}$ mu (paid in full).
   - Overheads: 6 months $\times 300 = \mathbf{1,800}$ mu. May and June (600 mu) unpaid $\implies$ **Overheads Payable** $= \mathbf{600}$ mu.
   - Renting (Apr–Jun, 3 months): $3 \times 400 = \mathbf{1,200}$ mu. Payable at 120 days $\implies$ **Renting Debt** $= \mathbf{1,200}$ mu.

4. **Depreciation & Amortization:**
   - Patent ($2,400/48$ mo $\times 3$ mo): $\mathbf{150}$ mu (Net BV $= 2,250$ mu).
   - Machine 1 ($500 - 100$ res val $= 400$ base over 4 yrs, semester dep): $\mathbf{50}$ mu (Accum dep $= 100$ mu). The 4-year life follows from the opening accumulated depreciation: 50 mu over the 6 months since 1 July 2014 gives 100 mu/year, so life $= 400/100 = 4$ years.
   - Machine 2 ($12,000/48$ mo $\times 5$ mo active): $\mathbf{1,250}$ mu (Accum dep $= 1,250$ mu).

- Total Depreciation in P&L: $150 + 50 + 1,250 = \mathbf{1,450}$ mu.

5. **Bank Credit Interest:** $2,000 \times 12\% \times \frac{5}{12} = \mathbf{100}$ mu (Accrued interest payable = 100 mu).
6. **EBIT, Tax & Net Income:**

$$\text{EBIT} = 23,280 - 14,400 - (3,000 + 1,800 + 1,200 + 1,450) = 8,880 - 7,450 = \mathbf{1,430}\ \text{mu}$$

$$\text{EBT} = 1,430 - 100 = \mathbf{1,330}\ \text{mu}$$

$$\text{Corporate Tax } (20\%) = 20\% \times 1,330 = \mathbf{266}\ \text{mu} \implies \text{Net Profit} = 1,330 - 266 = \mathbf{1,064}\ \text{mu}$$

Balanced Financial Statements at 30 June 2015

ASSETS (Activo)	Amount (mu)	LIABILITIES & EQUITY (Pasivo y PN)	Amount (mu)
NON-CURRENT ASSETS	**13,700**	**STOCKHOLDERS' EQUITY**	**16,114**
• Patent (Net Book Value: 2,400 - 150)	2,250	• Paid-in Capital (4,200 + 10,000)	14,200
• Machinery (Net BV: 12,500 - 1,350)	11,150	• Retained Earnings (2014)	850
• Long-term Customer Debt (July 2016)	300	• Net Profit for Semester	1,064
CURRENT ASSETS	**20,180**	**CURRENT LIABILITIES**	**17,766**
• Inventories (Stock)	0	• Short-term bank credit	2,000
• Accounts Receivable (June sales)	3,880	• Accrued loan interest	100
• Cash & Bank Deposits (Reconciled)	16,300	• Accounts payable (Suppliers Q2)	7,200
		• Debt for Machine 2 (due 1/7)	4,000
		• Debt for Patent (due 1/7)	2,400
		• Overheads payable (May & June)	600
		• Renting debt payable (Apr–Jun)	1,200
		• Taxes to pay (20% Corporate Tax)	266
TOTAL ASSETS	**33,880**	**TOTAL LIABILITIES & EQUITY**	**33,880**

Cash derivation (30/06/2015): inflows $1,200 + 1,500 + 19,400 + 2,000 + 10,000 = 34,100$ mu; outflows $350 + 450 + 4,800 + 3,000 + 1,200 + 8,000 = 17,800$ mu (2014 taxes, 2014 suppliers, Q1 purchases 400×12, personnel, overheads of January–April, Machine 2 down payment); cash $= 34,100 - 17,800 = \mathbf{16,300}$ mu.

EQUILIBRIUM VERIFICATION

$$\text{Assets } (\mathbf{33,880}\text{ mu}) \equiv \text{Liabilities } (\mathbf{17,766}\text{ mu}) + \text{Equity } (\mathbf{16,114}\text{ mu}) = \mathbf{33,880}\text{ mu}$$

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. Period 1 Jan to 30 Jun 2015 (6 months), accrual basis. Sales are uniform (200 u/month); the 30-day term leaves only June sales open; purchases are paid 90 days later, so Q1 purchases are settled by 30 June and all Q2 purchases remain open. Machine 1's 4-year life is inferred from the opening accumulated depreciation (50 over 6 months gives 100/year on a base of 500 - 100 = 400). Interest is simple (months/12); tax is 20% of EBT, left as "taxes to pay".

Phase 2: Definitions. $A = L + E$ with $E = \text{Paid-in capital} + \text{Retained earnings}_{2014} + \text{Net income}$. Assets carry debit balances; liabilities and equity carry credit balances; income-statement accounts are closed into equity.

Phase 3: Journal entries (m.u.). Each entry was generated and balanced in Python (debits = credits).

#	Event	Debit	Credit
1	Opening balance 31/12/2014 (A = L + E: 5,850 = 800 + 5,050)	Cash 1,200 Accounts receivable 1,800 Inventory 2,400 Machinery 500	Accum. depreciation 50 Taxes payable 350 Suppliers payable 450 Retained earnings 850 Paid-in capital 4,200
2	(i) Sales 1,200 u x 20 less 10% on the 30% VIP share	Accounts receivable 23,280 Sales discount 720	Sales 24,000
3	(i) Collection of Jan-May sales (30-day term; June stays open)	Cash 19,400	Accounts receivable 19,400
4	(ii) Purchases 1,000 u x 12	Inventory 12,000	Suppliers payable 12,000
5	(ii) Payment of Q1 purchases (400 u x 12, 90 days)	Suppliers payable 4,800	Cash 4,800
6	(iii) COGS = 2,400 opening + 12,000 purchases - 0 ending stock	COGS 14,400	Inventory 14,400
7	(iv) Payment of 2014 taxes and suppliers	Taxes payable 350 Suppliers payable 450	Cash 800
8	(v) Collection of 2014 receivables, 300 deferred to July 2016	Cash 1,500 Long-term receivable 300	Accounts receivable 1,800
9	(vi) Personnel + Social Security 6 x (350 + 150), paid in the month	Personnel 3,000	Cash 3,000
10	(vi) General expenses 6 x 300; Jan-Apr paid, May-Jun open	Overheads 1,800	Cash 1,200 Overheads payable 600
11	(vii) Computer renting 3 x 400 (term 120 days, nothing due yet)	Renting 1,200	Renting payable 1,200

#	Event	Debit	Credit
12	(viii) Patent purchased 1 April, payable 1 July	Patent 2,400	Patent payable 2,400
13	(viii) Patent amortization 2,400/48 x 3	Amortization 150	Accum. amortization 150
14	(ix) Bank credit received 1 February	Cash 2,000	Bank credit 2,000
15	(ix) Accrued interest 2,000 x 12% x 5/12	Interest expense 100	Accrued interest 100
16	(x) Depreciation machine 1: (500 - 100)/4 x 6/12	Depreciation 50	Accum. depreciation 50
17	(xi) Capital increase 1 February	Cash 10,000	Paid-in capital 10,000
18	(xi) Machine 2: 12,000 with 8,000 cash, 4,000 payable 1 July	Machinery 12,000	Cash 8,000 Machine payable 4,000
19	(xi) Depreciation machine 2: 12,000/48 x 5	Depreciation 1,250	Accum. depreciation 1,250
20	(xii) Corporate tax 20% x 1,330	Tax expense 266	Taxes payable 266

Closing ledger account	Balance (m.u.)
Cash	16,300 Dr
Accounts receivable	3,880 Dr
Long-term receivable	300 Dr
Inventory	0 Cr
Patent	2,400 Dr
Accum. amortization	150 Cr
Machinery	12,500 Dr
Accum. depreciation	1,350 Cr
Suppliers payable	7,200 Cr
Overheads payable	600 Cr
Renting payable	1,200 Cr
Patent payable	2,400 Cr
Machine payable	4,000 Cr
Bank credit	2,000 Cr
Accrued interest	100 Cr
Taxes payable	266 Cr
Paid-in capital	14,200 Cr
Retained earnings	850 Cr

Phase 4: Units and sanity checks. Currency: m.u. Accounting equation: $33{,}880 = 17{,}766 + 16{,}114$. Cash identity: $1{,}200 + 34{,}100 - 17{,}800 = 16{,}300$, equal to the ledger cash balance. Limit check: if no June sale were open, receivables would be 0 and cash 3,880 higher while profit would stay the same (accrual): receivables move cash, not profit. No official numerical key is available in `sources/` (only the statement PDF), so the figures stand on the recomputation above.

Exam Problem 2 • Multi-Year Horizon

INFOPYMES – Full Cycle Financial Statements, 2016–2017 & ROA/ROE

Original statement · T3_Additional_EXERCISES.pdf, p. 3

Three entrepreneurs founded **INFOPYMES** on 1 January 2016 with 100,000 € equity capital. A bank granted a 5-year loan of 100,000 € at 10% annual interest, returning 20,000 € principal each 31 December plus due interest. Computer equipment was bought on 01/01/2016 for 100,000 € (paid 01/04/2016; 5-year life, linear, 0 residual value). Online revenues: 300,000 €. Premises rent: 1,000 €/month. Permanent staff: 5,000 €/month. Utilities: 500 €/month. Variable consulting staff: 174,000 €. Results declared at 23:59 on 31 December; corporate tax 30% paid immediately; dividends 40% paid to partners.

1. Construct 2016 Income Statement.

2. Construct 31/12/2016 Balance Sheet at 24:00.

3. Construct 31/12/2017 Balance Sheet at 24:00 (assume no new investments, cash and retained earnings unchanged).

4. Determine ROA and ROE in 2017.

EXAM DAY STRATEGY: UNDERSTANDING "AT 24:00 AFTER PAYMENTS"

Because the prompt specifies that taxes, loan repayments, and dividends are paid *immediately before 24:00* on 31 December:

- No corporate taxes payable appear on the 2016 balance sheet.
- No dividends payable appear on the 2016 balance sheet.
- The bank loan principal drops immediately from €100,000 to €80,000.
- In 2017, the second repayment drops the loan principal from €80,000 to €60,000, while accumulated depreciation reaches €40,000.

Calculations & Financial Statements

1. **2016 Income Statement:**
   - Operating Expenses: Rent €12,000 + Permanent Staff €60,000 + Utilities €6,000 + Variable Wages €174,000 + Depreciation €20,000 (100k/5) = **€272,000**.
   - $\text{EBIT} = 300,000 - 272,000 = \mathbf{28,000}$ €.
   - Financial Interest (10% of €100,000): $\mathbf{10,000}$ €.
   - $\text{EBT} = 28,000 - 10,000 = \mathbf{18,000}$ €.
   - Tax (30%): $0.30 \times 18,000 = \mathbf{5,400}$ €.
   - Net Income $= \mathbf{12,600}$ € $\implies$ Dividends $(40\%) = \mathbf{5,040}$ €, Retained Earnings $(60\%) = \mathbf{7,560}$ €.

2. **Ending Cash at 31/12/2016:**

$$\text{Inflows } (100\text{k} + 100\text{k} + 300\text{k}) - \text{Outflows } (100\text{k} + 12\text{k} + 60\text{k} + 6\text{k} + 174\text{k} + 20\text{k} + 10\text{k} + 5.4\text{k} + 5.04\text{k}) = \mathbf{107,560}\ €$$

(In the outflows, the second 20k is the 31 December loan principal repayment, not depreciation.)

BALANCE SHEET COMPARISON	At 31/12/2016 (€)	At 31/12/2017 (€)	Account Interpretation
• Computer Equipment (Cost)	100,000	100,000	Fixed capital asset
• (-) Accumulated Depreciation	(20,000)	(40,000)	1 year vs 2 years linear wear
TOTAL NON-CURRENT ASSETS	**80,000**	**60,000**	**Net Book Value**
• Cash & Liquid Treasury	107,560	107,560	Stated unchanged by prompt
TOTAL ASSETS	**187,560**	**167,560**	**Balanced Asset Base**
• Subscribed Share Capital	100,000	100,000	Partners' equity
• Retained Earnings	7,560	7,560	Stated unchanged by prompt
TOTAL OWNERS' EQUITY	**107,560**	**107,560**	**Net Worth**
• Long-term Bank Debt	60,000	40,000	Remaining balance > 1 year
• Short-term Bank Debt	20,000	20,000	Principal due next 31 Dec
TOTAL LIABILITIES & EQUITY	**187,560**	**167,560**	**Perfect Match!**

Determination of ROA and ROE in 2017

$$\text{ROA}_{\mathbf{2017}} = \frac{\text{EBIT}_{2017}}{\text{Total Assets}_{2017}} = \frac{28,000\ €}{167,560\ €} \approx \mathbf{16.71\%}$$

$$\text{ROE}_{\mathbf{2017}} = \frac{\text{Net Income}_{2017}}{\text{Total Equity}_{2017}} = \frac{14,000\ €}{107,560\ €} \approx \mathbf{13.02\%}$$

(Where $\text{Net Income}_{2017} = (28,000 - 10\% \times 80,000) \times (1 - 0.30) = 20,000 \times 0.70 = 14,000\ €$*. This assumes that 2017 revenues and operating costs equal those of 2016 and that interest is* 10% *of the* $80,000$ *outstanding at the start of 2017. The statement also fixes retained earnings and cash at their 2016 values, so the 2017 net income of* $14,000$ *is not reflected as an increase in the balance sheet reserves.)*

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. 2016 is the first year; revenue is fully collected and all expenses are paid in the year. Interest is 10% of the balance at the start of the year (100,000 in 2016, 80,000 in 2017); 20,000 of principal is repaid each 31 December; tax (30% of EBT) and the 40% dividend are paid at 23:59 on 31 December, so no tax or dividend liability remains at 24:00. For 2017 the statement imposes unchanged cash and retained earnings and no new investment; operating results are assumed equal to 2016.

Phase 2: Definitions. $A = L + E$; ROA = EBIT/Total assets; ROE = Net income/Equity; equity = capital + retained earnings, with retained earnings = net income - dividends.

Phase 3: Journal entries 2016 (€). Each entry was balanced in Python (debits = credits).

#	Event	Debit	Credit
1	1 Jan: partners' capital	Cash 100,000	Capital 100,000
2	1 Jan: bank loan 5 years at 10%	Cash 100,000	Bank loan 100,000
3	1 Jan: computer equipment bought on credit (payable 1 Apr)	Equipment 100,000	Equipment payable 100,000
4	1 Apr: equipment supplier paid	Equipment payable 100,000	Cash 100,000
5	2016 consulting revenue, fully collected	Cash 300,000	Revenue 300,000
6	Rent 12 x 1,000	Rent 12,000	Cash 12,000
7	Permanent staff and Social Security 12 x 5,000	Staff 60,000	Cash 60,000
8	Electricity, phone 12 x 500	Utilities 6,000	Cash 6,000
9	Variable wages of consulting staff	Variable wages 174,000	Cash 174,000
10	Depreciation 100,000 / 5	Depreciation 20,000	Accum. depreciation 20,000
11	31 Dec: interest 10% x 100,000 paid	Interest 10,000	Cash 10,000
12	31 Dec: loan principal repayment	Bank loan 20,000	Cash 20,000
13	31 Dec: corporate tax 30% x 18,000 paid	Tax 5,400	Cash 5,400
14	31 Dec: dividends 40% x 12,600 paid	Dividends 5,040	Cash 5,040

Closing ledger account (24:00, 31/12/2016)	Balance (€)
Cash	107,560 Dr
Equipment	100,000 Dr
Accum. depreciation	20,000 Cr
Bank loan	80,000 Cr
Capital	100,000 Cr

Step-by-step lines: $\text{EBIT} = 300{,}000 - (12{,}000 + 60{,}000 + 6{,}000 + 174{,}000 + 20{,}000) = 28{,}000$; $\text{EBT} = 28{,}000 - 10{,}000 = 18{,}000$; tax $= 0.30 \times 18{,}000 = 5{,}400$; net income $= 12{,}600$; dividends $= 0.40 \times 12{,}600 = 5{,}040$; retained earnings $= 7{,}560$ (closed from income and dividends, so not a ledger line above). Balance 2016: $80{,}000 + 107{,}560 = 187{,}560 = 80{,}000 + 100{,}000 + 7{,}560$.

2017 bridge. Interest $= 0.10 \times 80{,}000 = 8{,}000$; EBT $= 20{,}000$; tax $= 6{,}000$; net income $= 14{,}000$. Cash after interest and tax $= 28{,}000 + 20{,}000 - 8{,}000 - 6{,}000 = 34{,}000$; minus principal 20,000 gives 14,000, which equals the dividend if the whole net income is distributed. Hence $\Delta\text{Cash} = 14{,}000 - 14{,}000 = 0$ and $\Delta\text{RE} = 14{,}000 - 14{,}000 = 0$, consistent with the statement. Ratios: $\text{ROA} = 28{,}000/167{,}560 = 16.71\%$; $\text{ROE} = 14{,}000/107{,}560 = 13.02\%$.

Phase 4: Units and sanity checks. Currency: euro (€); ratios are dimensionless. $A = L + E$: $187{,}560 = 80{,}000 + 107{,}560$ (2016) and $167{,}560 = 60{,}000 + 107{,}560$ (2017). Cash identity 2016: $500{,}000 - 392{,}440 = 107{,}560$. ROE is below ROA because interest and tax are deducted from its numerator, even though the equity base is smaller than total assets. No official numerical key is available in `sources/` (only the statement PDF).

Exam Problem 3 • Advanced Equity & Dividends

WIFINET S.L. – Telecommunications Accounting Cycle & Dividend Policies

Original statement · T3_Additional_EXERCISES.pdf, p. 4

Wifinet S.L. incorporated 1/1/2016 with 6,003 mu registered capital. In 2016 took a 2,000 mu loan at 5% (due 2019), bought equipment for 5,371 mu (20-year life, 26 mu accum dep in 2016), technology licenses for 206 mu (5-year life), and financial shares for 636 mu. 2016 reflected a loss of 340 mu and cash of 1,476 mu.
In 2017: 5,000 clients paying 0.032 mu/mo subscription fees, plus 46 mu online sales (receivables at end of year: 164 mu). Purchased 680 mu materials (560 mu paid immediately, rest deferred; half consumed in sales). Staff costs 559 mu paid in full. Outside services 312 mu paid in full. Administrative costs 300 mu (40% paid, 60% deferred). Corporate tax rate 30% payable Q1 2018. Retain all profits to offset 2016 losses.

1. Prepare 2017 Income Statement.

2. Construct 31/12/2017 Balance Sheet.

3. What changes if a 20% dividend is paid at 24:00 on 31/12/2017?

4. What changes if a 20% dividend is approved before year-end to be paid on 30 June 2018?

EXAM DAY STRATEGY: DIVIDEND PAYMENT VS DIVIDEND DECLARATION

- **Paid on 31 Dec (Cash Outflow):** Cash decreases by dividend amount; Retained Earnings decrease by dividend amount. Total Assets and Total Equity contract.
- **Payable in the Future (Accrued Debt):** No cash leaves the company in 2017! Retained Earnings decrease, but a current liability (**Dividends Payable**) is created. Total Assets remain unchanged!
- **Prior Year Loss:** The 340 mu loss sits in Equity as a negative line item, offset by 2017's net profit.

Calculations & Statements

- **Revenues:** $5,000 \times 0.032 \times 12 + 46 = 1,920 + 46 = \mathbf{1,966.00}$ mu.
- **COGS:** Half of $680 = \mathbf{340.00}$ mu (Inventory = 340.00 mu).
- **Depreciation:** Equipment $(5,371/20 = 268.55)$ + Licenses $(206/5 = 41.20) = \mathbf{309.75}$ mu.
- **Operating Expenses:** $340 + 559 + 312 + 300 + 309.75 = \mathbf{1,820.75}$ mu.
- **EBIT, EBT & Net Income:** $\text{EBIT} = 1,966 - 1,820.75 = 145.25$ mu. Interest $= 5\% \times 2,000 = 100$ mu. $\text{EBT} = 45.25$ mu. Tax $(30\%) = 13.575$ mu. Net Income $= \mathbf{31.675}$ mu.
- **Ending Cash:** $1,476 + 1,802$ (collections) $- 560 - 559 - 312 - 120 - 100 = \mathbf{1,627.00}$ mu.

ASSETS at 31/12/2017	Amount (mu)	LIABILITIES & EQUITY at 31/12/2017	Amount (mu)
• Licenses (Net BV: 206 - 41.20)	164.800	• Subscribed Share Capital	6,003.000

ASSETS at 31/12/2017	Amount (mu)	LIABILITIES & EQUITY at 31/12/2017	Amount (mu)
• Technical Equipment (Net: 5,371 - 294.55)	5,076.450	• (-) Prior Year Loss (2016)	(340.000)
• Financial Shares in other companies	636.000	• Net Income Year 2017	31.675
TOTAL NON-CURRENT ASSETS	**5,877.250**	**TOTAL OWNERS' EQUITY**	**5,694.675**
• Inventories (Supplies)	340.000	• Long-term Bank Loan (due 2019)	2,000.000
• Accounts Receivable	164.000	• Accounts Payable (Suppliers)	120.000
• Cash & Bank Deposits	1,627.000	• Other Creditors (Admin expenses)	180.000
		• Taxes Payable (due Q1 2018)	13.575
TOTAL ASSETS	**8,008.250**	**TOTAL LIABILITIES & EQUITY**	**8,008.250**

DIVIDEND POLICY COMPARISON (QUESTIONS 3 & 4)

20% Dividend Amount: $20\% \times 31.675 = \mathbf{6.335}$ mu.

- **If Paid on 31/12/2017:** Cash drops to $1,620.665$ mu. Equity drops to $5,688.340$ mu. Total Balance $= 8,001.915$ mu.
- **If Payable on 30/06/2018:** Cash remains $1,627.000$ mu. Current Liabilities increase by $+6.335$ mu (**Dividends Payable**), Equity drops to $5,688.340$ mu. Total Balance remains balanced at **8,008.250 mu**!

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. The 31/12/2016 position is reconstructed from the data: capital 6,003; loan 2,000; equipment 5,371 less 26 accumulated depreciation; licenses 206; financial shares 636; cash 1,476; loss -340. Check: $1,476 + 5,345 + 206 + 636 = 7,663 = 2,000 + 5,663$. Licenses are assumed not amortized in 2016 (the statement gives no 2016 amortization) and are amortized over 5 years in 2017; equipment is depreciated over 20 years. Stocks were exhausted at end 2016, so half of the 680 purchases is consumed (COGS 340). Interest is paid at year end; tax is payable in Q1 2018.

Phase 2: Definitions. $A = L + E$ with $E = \text{Share capital} - \text{Prior-year loss} + \text{Net income}_{2017}$. Financial shares are a non-current investment, not an expense. A declared dividend reduces equity at once but reduces cash only when paid; in between it is a current liability.

Phase 3: Journal entries 2017 (m.u.). Each entry was balanced in Python (debits = credits).

#	Event	Debit	Credit
1	Opening balance 31/12/2016 (A = 7,663 = L 2,000 + E 5,663)	Cash 1,476.000 Technical equipment 5,371.000 Licenses 206.000 Financial shares 636.000 Prior-year loss 340.000	Accum. depreciation 26.000 Bank loan 2,000.000 Share capital 6,003.000
2	Revenue: 5,000 x 0.032 x 12 + 46 online sales (164 uncollected)	Cash 1,802.000 Accounts receivable 164.000	Revenue 1,966.000

#	Event	Debit	Credit
3	Purchases of materials 680 (560 paid, 120 deferred)	Inventory 680.000	Cash 560.000 Accounts payable 120.000
4	Consumption: half of the purchases	COGS 340.000	Inventory 340.000
5	Staff costs paid	Staff 559.000	Cash 559.000
6	Supplies and outside services paid	Services 312.000	Cash 312.000
7	Administrative expenses 300 (40% paid, 60% deferred)	Administration 300.000	Cash 120.000 Other creditors 180.000
8	Depreciation equipment 5,371/20	Depreciation 268.550	Accum. depreciation 268.550
9	Amortization licenses 206/5	Amortization 41.200	Licenses 41.200
10	Interest 5% x 2,000 paid at year end	Interest 100.000	Cash 100.000
11	Corporate tax 30% x 45.25 (payable Q1 2018)	Tax expense 13.575	Taxes payable 13.575

Closing ledger account (31/12/2017)	Balance (m.u.)
Cash	1,627.000 Dr
Accounts receivable	164.000 Dr
Inventory	340.000 Dr
Technical equipment	5,371.000 Dr
Accum. depreciation	294.550 Cr
Licenses	164.800 Dr
Financial shares	636.000 Dr
Bank loan	2,000.000 Cr
Accounts payable	120.000 Cr
Other creditors	180.000 Cr
Taxes payable	13.575 Cr
Share capital	6,003.000 Cr
Prior-year loss	340.000 Dr

Income statement from the ledger: $1{,}966.000 - 340.000 = 1{,}626.000$; minus $1{,}480.750$ ($559.000 + 312.000 + 300.000 + 268.550 + 41.200$) gives EBIT $= 145.250$; minus interest 100.000 gives EBT $= 45.250$; minus tax 13.575 gives net income 31.675.

Requirement 3 (20% dividend paid at 24:00 on 31/12/2017): dividend $= 0.20 \times 31.675 = 6.335$. Entry: debit retained earnings 6.335, credit cash 6.335. Cash $1{,}627.000 - 6.335 = 1{,}620.665$; assets $8{,}001.915$; equity $5{,}688.340$; $5{,}688.340 + 2{,}313.575 = 8{,}001.915$. **Requirement 4** (decided in 2017, paid 30/06/2018): debit retained earnings 6.335, credit dividends payable 6.335. Assets unchanged ($8{,}008.250$); equity $5{,}688.340$; current liabilities $313.575 + 6.335 = 319.910$; total $5{,}688.340 + 2{,}000.000 + 319.910 = 8{,}008.250$.

Phase 4: Units and sanity checks. Currency: m.u. (as in the statement); three decimals are kept because 268.55, 41.20 and 13.575 are exact. $A = L + E$: $8{,}008.250 = 2{,}313.575 + 5{,}694.675$, and it still holds in both dividend variants. Cash identity: $1{,}476 + 1{,}802 - 560 - 559 - 312 - 120 - 100 = 1{,}627$. Limit check: paying at once shrinks both sides by 6.335; deferring only moves 6.335 from equity to liabilities. No official numerical key is available in `sources/` (only the statement PDF).

Exam Problem 4 • Industrial Manufacturing

CEMED – Industrial Manufacturing Accounting Cycle & Trade Discount Analysis

Original statement · T3_Additional_EXERCISES.pdf, p. 5

CERÁMICAS DEL MEDITERRÁNEO (CEMED) was created 1/1/2016 by 3 partners contributing 7 mu each in cash (total 21 mu).
• 1/2/2016: bought land for 10 mu (50% cash, 50% payable 1 year later); factory building for 6 mu in cash financed by a 10-year mortgage at 5% (20-year linear life, returns 10% principal each 31 Dec + interest).
• 1/4/2016: borrowed 5 mu at 6% from bank, returning 20% each semester for 2.5 years (1st repayment 1/10/2016, 2nd 1/4/2017). Purchased machinery for 3 mu (60-day payment delay; 10-year linear life).
• 1/7/2016: bought 150 tons raw material at 0.03 mu/ton (half used in sales; 60% paid at 90 days, 40% pending).
• Sales: 50 mu (70% collected). Personnel & SS: 0.8 mu paid in full. Tax: 30% paid on 15/01/2017. Reserves: 40% of net profits, 60% dividends payable on 31/12/2016.
1. Prepare 2016 Income Statement and Balance Sheet at 24:00 on 31/12/2016.
2. Analyze impact of 5% raw material purchase discount in exchange for paying the 60% in cash.

EXAM DAY STRATEGY: SEMIANNUAL VS ANNUAL LOAN CALCULATIONS

- **Factory Building (11 months):** $D = \frac{6.00}{20} \times \frac{11}{12} = 0.275$ mu. Mortgage interest: $6.00 \times 5\% \times \frac{11}{12} = 0.275$ mu. Principal returned on 31 Dec $= 0.60$ mu.
- **Bank Loan (Repayments & Accruals):** On 1 Oct returned 1.00 mu principal + paid 0.15 mu interest ($5 \times 6\% \times 0.5$). For Q4 (1 Oct to 31 Dec), remaining principal is 4.00 mu, accruing $4 \times 6\% \times \frac{3}{12} = \mathbf{0.06}$ mu unpaid interest payable in April 2017!
- **Machinery (9 months):** $D = \frac{3.00}{10} \times \frac{9}{12} = 0.225$ mu.

Statements & Balanced Verification

COGS $= 2.25$ mu. Operating Expenses $= 2.25 + 0.80 + 0.50$ (dep) $= 3.55$ mu. EBIT $= 50.00 - 3.55 =$ **46.45** mu. Financial Interest $= 0.275 + 0.210 =$ **0.485** mu. EBT $= 45.965$ mu. Tax (30%) $= 13.7895$ mu. Net Income $=$ **32.1755** mu (Reserves: 12.8702 mu, Dividends: 19.3053 mu).

ASSETS at 31/12/2016	Amount (mu)	LIABILITIES & EQUITY at 31/12/2016	Amount (mu)
• Land (Non-depreciable)	10.0000	• Subscribed Share Capital	21.0000
• Factory Building (Net: 6.00 - 0.275)	5.7250	• Retained Reserves (40% of Net Income)	12.8702
• Machinery (Net: 3.00 - 0.225)	2.7750	**TOTAL EQUITY**	**33.8702**

ASSETS at 31/12/2016	Amount (mu)	LIABILITIES & EQUITY at 31/12/2016	Amount (mu)
TOTAL NON-CURRENT ASSETS	**18.5000**	**• Long-term Mortgage (4.80) & Loan (2.00)**	**6.8000**
• Inventories (Raw Materials)	2.2500	• Land debt due 1/2/2017	5.0000
• Accounts Receivable (30% of sales)	15.0000	• Short-term Mortgage (0.60) & Loan (2.00)	2.6000
• Cash & Liquid Treasury	28.1697	• Accrued bank loan interest	0.0600
		• Accounts Payable (Raw materials 40%)	1.8000
		• Taxes Payable (due 15/01/2017)	13.7895
TOTAL ASSETS	**63.9197**	**TOTAL LIABILITIES & EQUITY**	**63.9197**

Cash derivation (31/12/2016): inflows $21 + 6 + 5 + 35 = 67.0000$ mu; outflows $5 + 6 + 3 + 2.7 + 0.8 + 0.6 + 0.275 + 1.0 + 0.15 + 19.3053 = 38.8303$ mu (land down payment, building, machinery, 60% of raw materials, personnel, mortgage principal and interest, bank loan principal and first-semester interest, dividends); cash $= 67.0000 - 38.8303 =$ **28.1697** mu.

TRADE DISCOUNT NEGOTIATION VERDICT (QUESTION 2)

A 5% discount on purchases reduces material cost by 0.225 mu. Because half was consumed, **COGS decreases by** 0.1125 mu, expanding EBIT by $+0.1125$ mu and Net Profit by $+0.1125 \times 0.70 = +\mathbf{0.07875}$ mu. Profit is strictly higher: the income statement does not record the time value of paying the 60% earlier, and both payment dates fall within 2016. The balance sheet also changes: inventories -0.1125 mu, accounts payable -0.09 mu, taxes payable $+0.03375$ mu, reserves $+0.0315$ mu, and year-end cash $+0.08775$ mu ($= 0.135$ saved on the supplier payment -0.04725 extra dividend), so that $\Delta\text{Assets} = \Delta(\text{Liabilities} + \text{Equity}) = -0.02475$ mu.

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. Calendar year 2016; partners' 21 paid in cash. The mortgage (6 at 5%) funds the building and its 10% principal plus annual interest are paid each 31 Dec. The bank loan (5 at 6%) is repaid 20% each semester from 1 Oct 2016, with interest on the amount actually due; interest for 1 Oct to 31 Dec ($4 \times 6\% \times 3/12 = 0.06$) is accrued. Machinery (3) is paid on 31 May. Raw materials (4.5) are bought on 1 Jul, half consumed; 60% paid on 1 Oct, 40% open at year end. Tax (30%) is payable 15 Jan 2017; dividends (60% of net income) are paid on 31 Dec and 40% is kept as reserves.

Phase 2: Definitions. $A = L + E$ with $E =$ Share capital + Reserves and Reserves $= 0.40 \times$ Net income. Land is not depreciated; building $6/20 \times 11/12$; machinery $3/10 \times 9/12$. Debt due within 12 months is current (mortgage 0.6; bank loan 2.0).

Phase 3: Journal entries, Part 1 (m.u.). Each entry was balanced in Python with exact fractions (debits = credits).

#	Event	Debit	Credit
1	1 Jan: partners' capital 3 x 7	Cash 21.0000	Share capital 21.0000
2	1 Feb: land 10 (50% cash, 50% due 1/2/2017)	Land 10.0000	Cash 5.0000 Land payable 5.0000
3	1 Feb: factory building 6 financed by mortgage at 5%	Building 6.0000	Mortgage 6.0000

#	Event	Debit	Credit
4	1 Apr: bank loan 5 at 6%	Cash 5.0000	Bank loan 5.0000
5	1 Apr: machinery 3, paid after 60 days (31 May)	Machinery 3.0000	Cash 3.0000
6	1 Jul: raw materials 150 t x 0.03 = 4.5	Inventory 4.5000	Suppliers payable 4.5000
7	Payment of 60% of the raw materials at 90 days (1 Oct)	Suppliers payable 2.7000	Cash 2.7000
8	Raw materials consumed: half	COGS 2.2500	Inventory 2.2500
9	Sales 50: 70% collected, 30% open	Cash 35.0000 Accounts receivable 15.0000	Sales 50.0000
10	Personnel and Social Security paid	Personnel 0.8000	Cash 0.8000
11	Depreciation building 6/20 x 11/12	Depreciation 0.2750	Accum. dep. building 0.2750
12	Depreciation machinery 3/10 x 9/12	Depreciation 0.2250	Accum. dep. machinery 0.2250
13	31 Dec: mortgage interest 6 x 5% x 11/12 and principal 10% x 6 paid	Interest 0.2750 Mortgage 0.6000	Cash 0.8750
14	1 Oct: loan principal 20% x 5 and interest 5 x 6% x 6/12 paid	Bank loan 1.0000 Interest 0.1500	Cash 1.1500
15	31 Dec: accrued interest 4 x 6% x 3/12 (paid 2/4/2017)	Interest 0.0600	Accrued interest 0.0600
16	31 Dec: corporate tax 30% (paid 15/1/2017)	Tax expense 13.7895	Taxes payable 13.7895
17	31 Dec: dividends 60% of net income paid	Dividends 19.3053	Cash 19.3053

Closing ledger account (31/12/2016)	Balance (m.u.)
Cash	28.1697 Dr
Accounts receivable	15.0000 Dr
Inventory	2.2500 Dr
Land	10.0000 Dr
Building	6.0000 Dr
Accum. dep. building	0.2750 Cr
Machinery	3.0000 Dr
Accum. dep. machinery	0.2250 Cr
Land payable	5.0000 Cr
Mortgage	5.4000 Cr
Bank loan	4.0000 Cr
Accrued interest	0.0600 Cr
Suppliers payable	1.8000 Cr
Taxes payable	13.7895 Cr
Share capital	21.0000 Cr

Statement lines: $\text{EBIT} = 50 - 2.25 - 0.8 - 0.5 = 46.45$; interest $= 0.275 + 0.15 + 0.06 = 0.485$; $\text{EBT} = 45.965$; tax $= 0.30 \times 45.965 = 13.7895$; net income $= 32.1755$; dividends $= 0.60 \times 32.1755 = 19.3053$; reserves $= 12.8702$ (closed from income and dividends, so not a ledger line above). Assets $18.5 + 45.4197 = 63.9197$; liabilities $5 + 5.4 + 4.0 + 0.06 + 1.8 + 13.7895 = 30.0495$ plus equity $21 + 12.8702 = 33.8702$.

Part 2 (5% discount, 60% paid in cash on 1 Jul): purchase price $4.5 \times 0.95 = 4.275$; payment 2.565. Changes with respect to Part 1:

Item	Change vs Part 1 (m.u.)
Net income	+0.07875
Cost of goods sold	-0.1125
Inventory	-0.1125
Cash	+0.08775
Suppliers payable	-0.09
Taxes payable	+0.03375
Total assets	-0.02475
Total liabilities + equity	-0.02475

Cash: smaller supplier payment $2.70 - 2.565 = +0.135$, minus the extra dividend 0.04725 (60% of 0.07875) gives $+0.08775$. Balance check: $\Delta A = -0.1125 + 0.08775 = -0.02475 = -0.09 + 0.03375 + 0.0315 = \Delta(L + E)$.

Phase 4: Units and sanity checks. Currency: m.u.; $A = L + E$: $63.9197 = 30.0495 + 33.8702$ (Part 1) and $63.89495 = 63.89495$ (Part 2). Cash identity Part 1: $67.0000 - 38.8303 = 28.1697$. Limit check: with a 0% discount every change vanishes; net income rises by the after-tax saving $0.1125 \times (1 - 0.30) = 0.07875$, so profit is higher with the discount. No official numerical key is available in `sources/` (only the statement PDF).